VUT

Vaal University of Technology

FACULTY:	**APPLIED AND COMPUTER SCIENCES**
DEPARTMENT:	**MATHEMATICS**
MODULE TITLE:	**MATHEMATICS 1**
MODULE CODE:	**AMATH 1A**
COMPILED BY:	**J. OWUSU-MENSAH**
YEAR:	**2018; SEMESTER 1**

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1966 - 2016

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1 MODULE INFORMATION

1.1 WORD OF WELCOME

The Department of Mathematics welcomes you as a learner to the Faculty of Applied and Computer Sciences at the Vaal University of Technology.

The department strives towards integration of existing knowledge with new knowledge and to afford the learners the ability to:

- think creatively and logically
- gain knowledge and skill of solving mathematical problems, and to apply techniques of solving problems to your related learning area of specialization, in the field of engineering.
- apply mathematical knowledge to make a positive contribution to the fields of your career once you have completed your studies.

NB: Visit the notice boards on the ground level of the E-block at least twice a week for important announcements and information.

1.2 PHILOSOPHY OF TEACHING AND LEARNING

1.2.1 DEPARTMENT'S view of Teaching and Learning

The Department of Mathematics' view of learning is that it is a process that involves cognitive, emotional, and environmental influences and experiences for acquiring, enhancing, or making changes in a learner's knowledge, skills, values and world views. Mathematics is a deductive science where learning starts with a few basic truths (axioms) and proceeds by deducing further truths in a logical way. Learning Mathematics can, thus, be compared to building a tower wherein the stability of each layer depends on both the foundation and the preceding layers. Individual learners bring their own unique experiences, memories and attitudes into the learning environment. Teaching is thus an opportunity to facilitate learning by formulating and effecting best practices that combine these individual perspectives to form knowledge.

1.2.2 Department's role in creating good learning contexts for students

A good learning context is characterized by four attributes namely; an appropriate motivational context, a well-structured knowledge base, relevant learner activity, formative feedback, and reflective practice and self-monitoring (Biggs & Tang, 2007). Lecturers in the Mathematics department will strive to create good learning contexts for students through strict adherence to seven principles of good practice (Chickering & Gamson, 1987) namely:

- Encouraging frequent contact between lecturers and students in and outside classes in order to keep students motivated. Lecturers will at all times be expected to conduct themselves in such a manner that students will view them as role models.

- Encouraging cooperation among students and staff. Good learning, like good work, is collaborative and social, not competitive and isolated. Learning is more enhanced when it is more like a team effort than a solo race.
- Encouraging active learning. Students do not learn much just sitting in classes listening to lecturers, memorizing pre-packaged assignments, and spitting out answers. Lectures must be centred on students actively solving problems themselves.
- Giving prompt feedback. Lecturers will give appropriate feedback on performance for students to reflect on what they have learnt as well as what they still need to learn.
- Emphasizing time on task. Learning to use one's time is critical for everyone, and time plus energy equals learning.
- Communicating high expectations. Expecting to succeed must be high priority for students and it is through this that they will strive to conquer their studies.
- Respecting diverse talents and ways of learning. Lecturers will give students opportunities to show their own talents and learn in ways that work for them and thereby helping others.

1.2.3 Students Responsibilities

To achieve meaningful learning, students are expected to complement the lecturers' role of creating good learning environment by:

- attending lectures and tutorials regularly
- being able to reflect on how well they are learning
- monitoring their progress
- modifying their learning strategies, where necessary, to improve their learning.
- actively participating in learning tasks (the learning of Mathematics is in the "**doing**" rather than being a spectator)
- observing punctuality at all times
- cultivating the love of learning
- collaborating with peers

Ideally, knowledge is not imparted by the lecturer, but it is created by learners themselves through their interaction and cooperation. The mind-set of the learners should not be focused on what the lecturer is doing but on what they, as learners, should best be doing because ultimately what they learn depends on how they go about learning.

REFERENCES

Biggs, J. & Tang, C. (2007) *Teaching for Quality Learning at University*. Third Edition. Society for Research into Higher Education & Open University Press. Mc-Graw Hill.

Chickering, A.W., Gamson, Z.F. (1987) Seven Principles for Good Practice in Undergraduate Education, The Wingspread Journal.

1.3 CONTACT PERSONS

Name	Designation	Office No.	Telephone No. (Area Code 016)
Dr K. Zimba	Senior Lecturer	E008-2	
Dr R.K. Mahlobo	Senior Lecturer	E016-2	950 9819
Mr M.J. Mbongwe	Head of Dept	F112	
Mrs A.J. van Ellewee	Senior Lecturer	E008-4	950 6603
Dr K. Okosun	Senior Lecturer	E016-1	950 9539
Dr B. Sibanda	Lecturer	E016-7	
Mrs C.R. Makhalemele	Lecturer	E008-1	950 6759
Mr T.T. Mthombeni	Lecturer	E008-3	950 6607
Dr J. Owusu-Mensah	Senior Lecturer	E016-5	950 6605
Mr G. Lekalakala	Lecturer	Secunda	017 6311971
Mr. M. Mukamuri	Lecturer	E016-3	950 6609
Mr M Ntshangase	Lecturer	E016-6	950 6608
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Maths centre		Y-block	

NB. Write down the name and the office number of your lecturer:

Name:_____ Office number:_____

1.4 RATIONALE

On completion of this module you should be able to understand and have knowledge of basic mathematical concepts and laws and know how to apply them. This module relates closely to all the other modules in the programme since it will improve your ability to understand the learning contents and enhance problem solving skills.

1.5 PREREQUISITES

A pass in Mathematics at Matric at level 4, or its equivalent

1.6 LEARNING MATERIAL

The content of this learner guide is the primary source of your learning material.

Additional learning material

Apart from this guide it is of utmost importance that you gain access to the following sources.

1.6.1. Prescribed Textbook:

A: Mathematics 1 Notes for Vaal University of Technology. Editors: J. Owusu-Mensah & C.R. Makhalemele

1.6.2 Reference Textbooks:

B: Basic Technical mathematics with Calculus (Ninth Edition) by Allyn J. Washington

C: Engineering Mathematics Through Applications (2nd edition) by Kuldeep Singh

D: Mathematics 1 for Technology Students by AM Badenhorst, CF Hitchcock and EHA Venter.

Only **non-programmable** calculators may be used in tests and examinations.

1.7 HOW TO STUDY

When you study Mathematics I (AMATH1E/S) you should:

- attentively study the outcomes on all levels (module, learning unit, learning section);
- view the module plan for a bird's eye view of what the module entails;
- view the list of contents of the first learning unit;
- page through the learning materials to determine the organization, difficulty level, and volume;
- read the learning material according to the instructions provided in the learner guide and learning outcomes;
- do all learning activities (exercises) in every learning section (or learning part) of the learner guide;
- be well prepared for all group meetings (discussions) according to the time schedule, in other words study the sections indicated/listed in the learner guide in order to take part in the group discussion;
- complete all self-evaluation exercises and mark each according to the answers/instructions/solutions provided;
- submit fully completed assignments/projects according to the time schedule;
- complete the summative evaluation [exercise(s)/questions] and follow the instructions provided.

Personalise the learning process through:

- giving subject specific learning tips from your own experience and suggest the use of different learning strategies to try and accommodate as many learning styles as possible.
- guiding learners in the application of different learning strategies that produce good subject specific results in achieving outcomes, like to identify and summarize core ideas and to use it.

1.8 ASSESSMENT

1.8.1 Semester and class tests

- Assessment will include two semester tests, a class test and some additional work. All marks will be converted to percentages; then the mark for the class test will be multiplied by 0,2, the mark for each semester test by 0,35 and the mark for additional work (AW) by 0,1 to obtain the semester mark.
- $\boldsymbol{Semester\ mark} = (\boldsymbol{CT} \times \mathbf{0.2}) + (\boldsymbol{ST1} \times \mathbf{0.35}) + (\boldsymbol{ST2} \times \mathbf{0.35}) + (\boldsymbol{AW} \times \mathbf{0.1})$
- Students who obtain a semester mark of 50% or more qualify to write the **FIRST OPPORTUNITY ASSESSMENT.**
- The additional work could consist of class work, homework, quizzes, assignments, a class test etc. as agreed between individual lecturers and their students.
- **All tests and additional work must be done in your OWN group with your OWN LECTURER.** Students **must** write semester tests in their assigned venues and indicate their lecturer's name on their test scripts. Lecturers will not take responsibility for tests that are lost and tests written in wrong venues will not be marked.

1.8.2 Examination

- A two hour exam paper shall be written for each of the two assessment opportunities.
- The final module mark is calculated as the average of the semester mark and the exam mark. To pass the module a student needs to obtain 40% (sub-minimum) or more in the examination and obtain a final module mark of 50% or more.
- Those who do not pass the module after the first opportunity assessment will automatically qualify to write the SECOND OPPORTUNITYASSESSMENT regardless of how much they would have obtained in the first opportunity.

1.8.3 Responsibilities of students

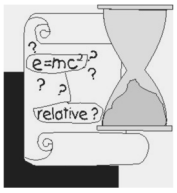
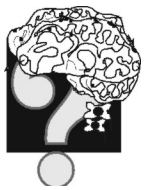
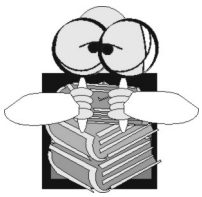
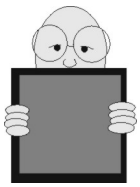
- Students must not write tests in pencil or erasable pen, use tippex or write answers next to each other on one page. If they do, they will not be allowed to query their marks.

- After every test, ensure that your marks are entered on the ITS and are correct. If not, inform your lecturer immediately. After the last test, revisit the ITS to double check whether all tests are recorded and are correct.
- Students are responsible for checking that marks are entered correctly in the ITS and that semester/year marks are calculated correctly in the ITS. Mistakes cannot be rectified later.
- Students must keep all tests, quizzes and class work as proof of marks earned.

1.8.4 Absence from a Test

- No excuse/reason shall be entertained for missing a test other than sickness or death of a close relative or family member.
- Valid documentation (medical certificate or death certificate) must be submitted immediately to the lecturer concerned in order to be allowed to write a special test.

1.8 ICONS USED IN THIS MODULE

1 	2	3	4
Estimated Study Time	Outcomes	Study the material in the prescribed book or article etc.	Individual Exercises
5 	6	7 	8 
Test your knowledge	Answers	Revision	General Survey

1.10 ACTION VERBS

In preparation for your studies in this module the action verbs used in the assessment of your work in this module are given to you in advance. Make sure that you understand the meaning of each and that you deliver your work accordingly.

- **Apply**
  Being able to apply theory in other situations. [*Example: Apply differentiation to solve a maximizing problem*]

- **Define**
  Explain the accurate meaning of a concept. [*Example: Define a differential equation.*]
- **Name/mention/list/state**
  Briefly name/mention/list/state without giving details. Neither a discussion nor an explanation is necessary. [*Example: State the exactness criteria.*]
- **Indicate**
  Point out, make known, state briefly. [*Example: Indicate the co-ordinates of the turning points*]
- **Prove**
  Show that a statement is true. [Example: *Prove the second shifting property*]
- **Derive**
  Prove a rule or property by means of logical reasoning. [*Example: Derive the general formula used to find Laplace transforms of derivatives.*]
- **Describe**
  Say exactly what something is like, give an account of the characteristics or nature of something, and explain how something works. No opinion or argument is needed. [*Example: Describe the characteristics of a Fourier series.*]
- **Illustrate**
  Draw a diagram, graph or sketch to show your knowledge of a concept or theorem. [*Example: Illustrate the continuity of a function by sketching the graph of the function.*]
- **Calculate/Determine**
  Obtain an answer to an operation. [*Example: Calculate:* $\sin 30^0 + \cos \pi$]
- **Represent (visually)**
  Describe a mathematical concept in different forms. [*Example: Give the graphical representation of the complex number* $z = x + yj$]
- **Solve**
  Find a solution to a given problem or equation. [*Example: Solve for x in the following equation:* $x: x^{lnx} = 4$.]
- **Distinguish**
  Indicate the difference between the characteristics of given data *[Example: Distinguish between the maximum and minimum points.*
- **Test**
  Compare results against given standardized values. [*Example: Test whether there is a significant difference between the two groups]*
- **Sketch**
  Draw a graph, indicate the critical points only. [*Sketch the graph of* $f(x) = \frac{1}{2-x^2}$]
- **Draw**
  Draw a graph to scale.
  [Example: Draw the graph of $f(x) = \begin{cases} x^2; 0 < x < 2 \\ -x + 4; 2 < x < 4 \end{cases}$ $and\ f(x+4) = f(x)$]
- **Apply**
  Use the theory or concept in another situation such as using a learned concept in a real life situation. [Example: Apply the knowledge of differential equations to solve the differential equation for the population growth model.]

1.11 MODULE PLAN

Learning unit 1	Binomial expansion, radian measure and limits of functions
Learning unit 2	Differentiation techniques
Learning unit 3	Integration techniques
Learning unit 4	Vectors
Learning unit 5	Complex numbers

AMATH1E: GUIDELINE FOR WEEKLY PLANNING OFCURRICULUM.-YEAR 2018

Lesson	Subject		Exercise
1	Revision: exponents, logs,		
2	Revision: Concept & evaluation of functions, Inverse functions		
3-4	Binomial theorem		
5	Radian measure		
6-7	Applications of radian measure		
8	Limits of functions		
9	Differentiation from first principles		
10	Derivatives of polynomials & product rule		
11	The quotient and chain rules		
12	Derivatives of trig functions		
13	Derivatives of exponential & log functions		
14	Higher order derivatives		
15	Implicit differentiation		
16	Logarithmic differentiation		
17-19	Applications		
20	Integration (Indefinite integrals)		
21	Definite integrals		
22	Area enclosed by two curves.		
23	Simpson's rule		
24	Rep.& magn of vectors. Resolving vectors		
25	Unit vectors and direction vectors.		
26	Scalar multiplication, addition and sub.		
27-28	Dot product, angle between two vectors and work done		
29-30	Determinant of a 2 x 2 matrix. Cross product and the moment of a vector		
31-32	Rep. of complex numbers and operations		
33	Equality of complex numbers		
34	Argand diagram, polar form & De Moivre's		
35	Calculating roots		

1.13 TIME SCHEDULE

You must make sure of all the dates of classes, tutorials, semester, class and special test, etc. and fill it in on the time schedule provided beneath.

Time schedule 2018 Semester 1							
Date	**Week**	**Lesson***		**Date**	**Week**	**Lesson***	
5 Feb	1	Admin		2 April	9	21	
						22	
						23	
12 Feb	2	1				Tut 7 & 8	
		2					
		3		9 April	10	24	
		Tut 1				25	
						26	
19 Feb	3	4					
		5					
		6		16 April	11	27	
		Tut 2				28	
						29	
26 Feb	4	7				Tut 9	
Class Test		**8**					
		9		23 April			
		Tut 3			12	30	
						31	
5 March	5	10		Sem Test 2		Tut 10	
		11					
		12		30 April		32	
		Tut 4			13		
12 March	6	13					
		14					
		15		2 May	14	33	
		Tut 5				34	
						35	
19 March	7	16				Tut 11	
				7 May	15		
Sem Test 1		18					
		Tut 6				Tut 12	
26 March	8	19		11 May			Year Marks
		20		18 May: 1st Opp			**Exams**

2 LEARNING UNIT INFORMATION

2.1 MODULE OUTCOMES

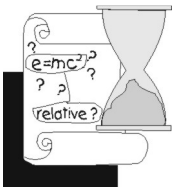
The qualifying learner should be able to:

- manipulate standard functions and techniques
- perform and apply differentiation techniques
- perform and apply integration techniques
- use, analyse and apply vectors to solve problems
- manipulate complex numbers

1LEARNING UNIT 1

LEARNING OUTCOME 1

STANDARD FUNCTIONS AND TECHNIQUES



Estimated study time

3 weeks

Intended Learning Outcomes

After completing this learning unit you should be able to:

- apply the laws of exponents and logarithms (Revision)
- evaluate functions and identify the domain and range of a given function (revision).
- apply trigonometric ratios and identities (Revision)
- raise a binomial to any power using Binomial theorem
- determine a given term in a binomial expansion
- convert angles from degrees to radians and vice versa
- apply radian measure to calculate arc length and areas of sectors and segments

CONTENTS

1.1 Revision:

Students are to do revision on the following topics. These topics will not be part of the assessment.

1.1.1 Exponents and logarithms

1.1.2 Concepts of Functions – domain, range, onto, one-to-one, many-to-one, inverse functions,

1.1.3 Trigonometric ratios, identities and inverse trigonometric functions, sine function

1.2 Binomial expansion

1.3 Radians

1.3.1 Arc length, area of sector and segment

1.4 Limits of functions

1.2 Binomial Expansion

Study the indicated material in the prescribed textbook (Maths 1 notes).

Study concepts on binomial expansion and do examples. p 4-11

Do exercise 1.3

Test your knowledge:

Expand as far as the term in x^3: $(1+2x)^{-1}$

Answer: $1-2x+4x^2-8x^3+\ldots$

1.3 Radians

Study the indicated material in the prescribed textbook (Maths 1 notes). p 28-30.

Study radian measure and changing angles from one form to the other.

Do exercise 1.5.2

1.3.1 Arc length, area of sector and segment

Study the indicated material in the prescribed textbook (Maths 1 notes). p 30-33

Know how to find area of a sector and segment

$Area\ of\ sector(A_{st}) = \frac{\theta}{2}r^2$ $\qquad$ $Area\ of\ segment(A_{sm}) = \frac{1}{2}r^2[\theta - \sin\theta]$

Do exercise 1.5.3

D**o** exercise 1.1.4

UNIT 1: Tutorial Questions

Tutorial 1

1.1 Write down the binomial expansion of $(x+2y)^4$ and use your expansion to evaluate $(1{,}02)^4$ correct to 3 decimal places

1.2 Use the Binomial Theorem to expand each binomial

1.2.1 $(x-1)^5$ 1.2.2 $(2x+3y)^4$

1.2.3 $(\frac{x}{2}+2)^7$ 1.2.4 $(\frac{1}{x}+y)^3$

1.3 Determine the first four terms of the given expansion:

1.3.1 $\sqrt{1+x}$ 1.3.2 $(1-3x)^{-2}$ 1.3.3 $\sqrt[3]{1-x}$

1.4 Find, without expanding, the specified term:

1.4.1 the term containing x^7 of $(x+y)^{10}$

1.4.2 the term containing x^3 of $(x-2)^{12}$

1.4.3 the term containing x^9 of $(5x-2y)^{11}$

1.4.4 the term containing $\frac{1}{x^2}$ of $\left(\frac{a}{x}+x\right)^6$

1.5 Find the middle term in the expansion of the following:

1.5.1 $(\sqrt{x}-\frac{2}{x^2})^{10}$ 1.5.2 $(1-x^2y^{-3})^{12}$

1.6 Determine the indicated term in the expansion of:

1.6.1 5th term of $(2x-3b)^{12}$ 1.6.2 6th term of $\left(b^2+\frac{1}{2b}\right)^{20}$

ANSWERS

1.1. $(x+2y)^4 = x^4+8x^3y+24x^2y^2+32xy^3+16y^4$. $(1{,}02)^4 = 1.082$

1.2. Expansion using binomial theorem:

1.2.1. $(x-1)^5 = x^5-5x^4+10x^3-10x^2+5x-1$

1.2.2. $(2x+3y)^4 = 16x^4+96x^3y+216x^2y^2+216xy^3+81y^4$

1.2.3. $(\frac{x}{2}+2)^7 = \frac{1}{128}x^7+\frac{7}{32}x^6+\frac{21}{8}x^5+\frac{35}{2}x^4+70x^3+168x^2+224x+128$

1.2.4. $(\frac{1}{x}+y)^3 = \frac{1}{x^3}+\frac{3y}{x^2}+\frac{3y^2}{x}+y^3$

1.3. Finding first four terms:

1.3.1. $\sqrt{1+x} = 1+\frac{1}{2}x-\frac{1}{8}x^2+\frac{1}{16}x^3-\ldots$

1.3.2. $(1-3x)^{-2} = 1+6x+27x^2+108x^3-\ldots$

1.3.3. $\sqrt[3]{1-x} = 1-\frac{1}{3}x-\frac{1}{9}x^2-\frac{5}{81}x^3-\cdots$

1.4. Finding a specific term:

1.4.1. Term containing x^7 of $(x+y)^{10} = 120x^7y^3$

1.4.2. Term containing x^3 of $(x-2)^{12} = -112\ 640\,x^3$

1.4.3. Term containing x^9 of $(5x-2y)^{11} = 429687500x^9y^2$

1.4.4. Term containing $\frac{1}{x^2}$ of $\left(\frac{a}{x}+x\right)^6 = 15a^4(\frac{1}{x^2})$

1.5. Finding the middle term:

1.5.1. Mid-term of $(\sqrt{x}-\frac{2}{x^2})^{10} = -\frac{8064}{x^{7.5}}$

1.5.2. Mid-term of $(1-x^2y^{-3})^{12} = 924\frac{x^{12}}{y^{18}}$

1.6. Determining specified terms.

1.6.1. 5th term of $(2x-3b)^{12} = 10264320\ x^8b^4$

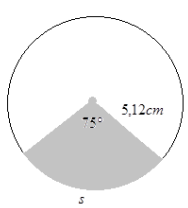
1.6.2. 6th term of $\left(b^2+\frac{1}{2b}\right)^{20} = 484{,}5b^{25}$

Tutorial 2

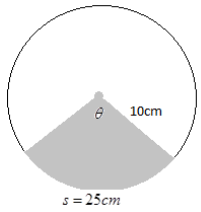
2.1. Convert to radian measure: a) $70°$ $b)$ $115°$ $c)$ $124°$

2.2. Convert the following angles in radian measure to degrees: $\frac{\pi}{3}; 1; 32; 5$

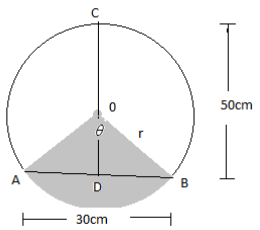
2.3. a) The radius of a circle is 5.12cm and a centre angle $\theta = 75°$. Calculate the length s of the arc that subtends the angle, correct to 3 decimal places.



b) A circle has a radius of 10cm and an arc length of 25cm. Calculate the area A_{ST} of the sector enclosed by the radii.



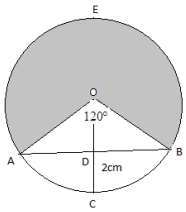
c) In the circle below, the chord AB is 30 cm and the line CD through centre O is 50 cm.



Calculate

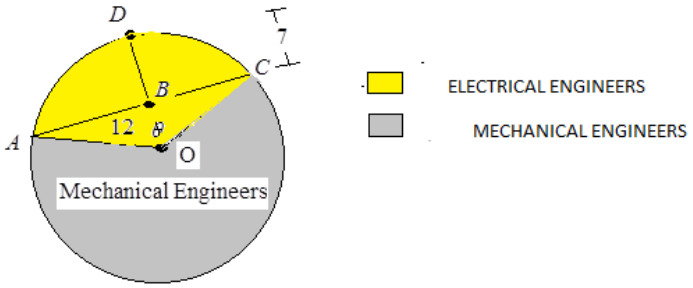
(i) The radius of the circle
(ii) The angle at the centre (correct to 3 d.p.)
(iii) The area of the shaded part (correct to 3 d.p.)

2.4. Given circle centre O with $\theta = 120^{\circ}$ and DC = 2 cm, determine:



a) r , the radius of the circle.
b) the length of arc AEB.
c) the shaded area.

2.5 A pie-chart is used to represent the proportion of 400 mechanical and electrical engineers in a certain engineering company. Given the pie with centre O and central angle $\theta,\ r = 12\ units$ and $DB = 7\ units$ as shown below, determine the approximate number of mechanical engineers in the company. Show all your working.



ANSWERS

2.1a) $70° = 1{,}221730476$ $b)\,115° = 2{,}00712864$ $c)\ 124° = 2{,}164208272$

2.2.a) $\frac{\pi}{3} = 60°$ b) $1rad = \frac{180}{\pi}° = 57{,}29577951°$ $c)\ 32rad = 1833{,}464944°$ d) $5rad = 286{,}4788976°$

2.3a) $s = 6{,}702cm$ $b) A_{ST} = 125cm^2$ $c)(i) r = 27{,}25cm, (ii)\theta = 1{,}166, (iii) A_{st} = 432{,}914cm^2$

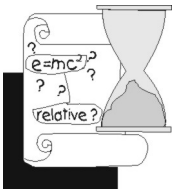
2.4a) $r = 4cm$, (b) $s = 16{,}7551608$ cm, (c) A $= 33{,}5096$cm ^{2}

2.5a) $\theta = 2,\ b)\ r_1 = 7{,}5cm,\ \ c)\ \ S_2 = 10cm$ 2.6 255

LEARNING UNIT 2

LEARNING OUTCOME 2

DIFFERENTIATION



Approximate study time

Four weeks

Intended Learning Outcomes

After completing this learning unit you should be able to:

- determine the derivative of a function using the limit definition
- find the derivative of a function using basic differentiation rules (sum, difference, product, quotient and chain rules)
- determine higher order derivatives
- apply logarithmic differentiation to differentiate products, quotients and function raised to the power a function
- find the derivative of implicit functions
- apply differentiation techniques to sketch the graph of a function by determining local maxima and minima (including rational functions)
- apply differentiation to solve velocity and acceleration problems for linear and circular motion
- approximate roots of equations using the Newton-Raphson method

CONTENTS

2.1 Limit of a function
2.1.1 The derivative from first principles
2.1.2 Derivatives of polynomials
2.1.3 Rules of differentiation (sum and difference, product, quotient and chain rules)
2.1.4 Derivatives of trigonometric functions
2.1.5 Derivatives of exponential and logarithmic functions
2.1.6 Higher order derivatives
2.1.7 Implicit differentiation and logarithmic differentiation
2.2 Applications of differentiation
2.2.1 Curve sketching (polynomials & rational functions only, local maximum and minimum
2.2.2 Velocity and acceleration for linear and circular motion
2.2.3 Approximation of roots of equations using Newton-Raphson Method.

Study the indicated material in the prescribed textbook (Maths 1 notes).

Study the introductory remarks and do examples. p 18-21

Study the concept of how to use algebra to find limits.

Example: Evaluate $\lim_{x \to 9} \left(\frac{9-x}{3-\sqrt{x}} \right)$

2.1 Derivatives

2.1.1 Differentiation from first principles

Study the indicated material in the prescribed textbook (Maths 1 notes). p 42

Study the concept of differentiation from first principles. p 42

Do exercise 2.2 on p 43

2.1.2 Derivatives of polynomials

Study the section on derivatives of polynomials, including sum and difference. p 44

Do examples from the prescribed textbook.

Do exercise 2.3.2

2.1.3 Rules of differentiation (product, quotient and chain rules)

Study the indicated material in the prescribed textbook (Maths 1 notes). p 45-48

Study and know product, quotient and chain rules and do examples.

Study section on application of the chain rule.

Exercise 2.3.3, 2.3.4 & 2.3.5

2.1.4 Derivatives of trigonometric functions

Study the indicated material in the prescribed textbook (Maths 1 notes). p 48-49

Do exercise 2.4

2.1.5 Derivatives of exponential and logarithmic functions

Study the indicated material in the prescribed textbook (Maths 1 notes). p 50

Do exercise 2.5

2.1.6 Higher Order Derivatives

Know how to determine higher order derivatives and do examples. p 51

`-- Do exercise 2.6

2.1.7 Implicit differentiation and logarithmic differentiation

Study the indicated material in the prescribed textbook (Maths 1 notes). p 52-55
Study concepts under implicit differentiation and related examples
Know how to apply logarithmic differentiation to turn functions into simpler forms and do related examples.

Exercise 2.7 and 2.8

2.2 Applications of Differentiation

2.2.1 Curve Sketching (polynomials, rational functions, maxima and minima)

Study the indicated material in the prescribed textbook (Maths 1 notes). p56.
Study the introduction which describes the terminology used and examples.

Individual exercise: Exercise 2.9.1

2.2.2 Velocity and acceleration for linear and circular motion

Study the indicated material in the prescribed textbook (Maths 1 notes) p **59**

Exercise 2.9.2

2.2.3 Approximation of roots of equations using Newton –Raphson Method

Study the indicated material in the prescribed text book (Maths 1 notes). p 62

Know the Newton-Raphson formula and why it is needed for approximating roots.

Apply the formula to approximate roots of various functions and do examples.

Exercise 2.9.3

UNIT 2: Tutorial questions

Tutorial 3

1 Find the derivatives of the following from first principles:

1.1 $f(x) = -2x^3$ 1.2 $f(x) = \frac{2}{x}$ 1.3 $f(x) = \sqrt{x}$ 1.4 $f(x) = \frac{-3}{x^2}$

1.5 $f(x) = \frac{1}{\sqrt{x}}$ 1.6 $f(x) = (x+2)^2$ 1.7 $f(x) = x^2 + 2x$

Evaluate the following:

2.1 $\lim_{x\to 1}\left(\frac{x^3-1}{x-1}\right)$ 2.2 $\lim_{x\to 2}\left(\frac{x^2-7x+10}{x^2-4}\right)$ 2.3 $\lim_{x\to 9}\left(\frac{9-x}{3-\sqrt{x}}\right)$ 2.4 $\lim_{h\to 0}\left(\frac{\sqrt{x+h}-\sqrt{x}}{h}\right)$

2.5. $\lim_{x\to\infty}\frac{x-1}{7x+4}$ 2.6. $\lim_{x\to\infty}\frac{3x^2+4,5}{x^2-1,5}$ 2.7. $\lim_{t\to\infty}\frac{\sqrt{t^2+16}}{t+1}$ 2.8 $\lim_{t\to\infty}\frac{t^2+5}{\sqrt{64t^4+1}}$

Tutorial 4

1. Differentiate the following:

1.1 $y = \frac{2\sqrt{x}+1}{x^2}$ 1.2 $y = \frac{1}{x} + \sqrt{x^3}$ 1.3 $y = \frac{2}{x^2} - \frac{3}{x^3}$ 1.4 $y = 4\sin\theta - \cos\theta$

1.5 $y = \frac{1}{4}\sin x + \frac{\cos x}{2}$ 1.6 $y = e^x - e^{-x}$ 1.7 $y = \frac{2}{5}e^{-x}$ 1.8 $y = a^{f(x)}$

1.9 $y = 2^{x^2}$ 1.10 $y = \sqrt{x} + \log x$ 1.11 $y = 3\ln t + \frac{1}{2}e^{-2t}$

2. Chain, product and quotient rules: Determine the derivatives of each of the following:

2.1 $y = (3x^2+5)^{-2}$ 2.2 $y = sin^2 5x + cosec\ 3x$ 2.3 $y = 2x^2(5x-3)^3$

2.4 $y = 6x(3x^2-2x)^2$ 2.5 $y = \tan x - \sec x$ 2.6 $y = \frac{6x^2}{3-2x}$

2.7 $y = \frac{e^{2x}}{\sin 2x}$ 2.8 $y = e^{\pi x}\cos 2\pi x$ 2.9 $y = 5x\sin 2x + e^{(2x-1)}$

Tutorial 5

1. Implicit differentiation: Determine $\frac{dy}{dx}$ in each of the following equations:

1.1 $y^2 - y = x^2 + 4x$ 1.2 $xy^2 + 3y + x^2 = 2\pi^2$

1.3 $x^2y + \cos(xy) - 5y + x = 0$ 1.4 $y - 3xy - x\ln 5 = 0$

2.1 Determine all the higher order derivatives of each of the given functions

a) $y = x^3 + 5x^2 - 3x$ b) $y = 3x - x^4$ c) $y = (1-2x)^4$

2.2 Determine the second derivatives

a) $y = 2x - \sqrt{x}$ b) $x = \sin 5t + \cos 5t$ c) $y = x^2 \ln(x-1)$

2.3 Logarithmic differentiation: Determine $\frac{dy}{dx}$ in each case:

2.3.1 $y = \ln\left(\frac{e^x x^3 \cos x}{\sqrt{x}(x^2+5)}\right)$ 2.3.2 $y = \frac{\sqrt{x}}{(2x+1)^4 sin^2 3x}$

2.3.3 $y = (\cos x)^{\sin x}$ 2.3.4 $y = x^{(1+\ln x)}$

Tutorial 6

1.1 Determine and distinguish the turning point(s) for the functions. Sketch the graph.

a) $f(x) = 2 + 6x - 3x^2$ b) $f(x) = 2x^3 + 6x^2 - 5$ $c)$ $f(x) = x^4 - 6x^2$

1.2 a) An object is thrown into the air such that its height, h metres after t seconds is $h = 36t - 4t^2$, determine

i) the velocity at $t = 2$

ii) the time taken to reach the maximum height

b) A is a fixed point on a straight line AB. The displacement from A of a particle P moving on this line is given by $s = t^3 - 6t^2 + 9t$ at any time t. Determine an expression for the velocity P at time t, and hence using metres and seconds, determine:

i) the initial velocity

ii) the velocity when $t = 2$

iii) the times when P is momentarily at rest.

1.3 Use the Newton Raphson method to find the roots of the following equations closest to the values given (in brackets). Apply the rule until two successive approximations agree to 3 decimal digits.

a) $f(x) = x^3 - 3x + 1$ $(0,4)$ b) $f(x) = x^3 - 3x^2 - 2x + 3$ $(0,5)$

1.4 The velocity flow, v, of a liquid along a channel satisfies $v^3 - 6v^2 - 348v + 3112 = 0$. Given that there is a root of this equation between $v = 10$ and $v = 11$, find this root correct to 3 decimal places.

2. When viewed through a microscope a bacterium is seen to move in accordance with the equation $s = (3t + 4t^2) \times 10^{-5}$. Determine the velocity after 2 seconds.

3. The displacement, x, of a particle in simple harmonic motion is given by $x = \cos 3t - \sin 3t$. Show that $\frac{d^2x}{dt^2} + 9x = 0$.

4. The pressure, P, and volume, V, of a gas in a cylinder are related by $PV^n = C$ (where C is constant). Show that $\frac{d^2P}{dV^2} = \frac{n(n+1)P}{V^2}$.

5. Faraday's law states that the electromotive force, E, induced by N turns of coil with a flux, φ, passing through it is given by $E = -N\frac{d\varphi}{dt}$. If $\varphi = K\sin(2\pi f t)$ where K and f are constants, determine E.

6. The distribution function, F, of a set of components is given by $F = 1 - \frac{15}{2+\frac{t}{3}}$ where t is in years. Determine the density function $\frac{dF}{dt}$.

ANSWERS

Tutorial 3

1.1. $-6x^2$ 1.2. $\frac{-2}{x^2}$ 1.3. $\frac{1}{2\sqrt{x}}$ 1.4. $\frac{6}{x^3}$ 1.5. $\frac{-1}{2x^{1.5}}$ 1.6. $2x+4$ 1.7. $2x+2$

2.1. 3 2.2. $-\frac{3}{4}$ 2.3. 6 2.4. $\frac{1}{2\sqrt{x}}$ 2.5. $\frac{1}{7}$ 2.6. 3 2.7. 1 2.8. $\frac{1}{8}$

Tutorial 4

1.1 $\frac{-3}{x^{2.5}} - \frac{2}{x^3}$ 1.2 $\frac{-1}{x^2} + \frac{3}{2}x^{\frac{1}{2}}$ 1.3 $\frac{-4}{x^3} + \frac{9}{x^4}$ 1.4 $4\cos\theta + \sin\theta$

1.5 $\frac{1}{4}\cos x - \frac{1}{2}\sin x$ 1.6 $e^x + e^{-x}$ 1.7 $\frac{-2}{5}e^{-x}$ 1.8 $f'(x)a^{f(x)}\ln a$

1.9 $2x\left(2^{x^2}\right)\ln 2$ 1.10 $\frac{1}{2}x^{-0.5} + \frac{1}{x\ln 10}$ 1.11 $\frac{3}{t} - e^{-2t}$

2.1 $-12x(3x^2+5)^{-3}$ 2.2 $5\sin 10x - 3cosec\ 3x\cot 3x$ 2.3 $2x(5x-3)^2(25x-6)$

2.4 $18x^2(3x-2)(5x-2)$ 2.5 $\sec x\,(\sec x - \tan x)$ 2.6 $\frac{12x(3-x)}{(3-2x)^2}$

2.7 $\frac{2e^{2x}(\sin 2x - \cos 2x)}{(\sin 2x)^2}$ 2.8 $\pi e^{\pi x}(\cos 2\pi x - 2\sin 2\pi x)$ 2.9 $5(\sin 2x + 2x\cos 2x) + 2e^{2x-1}$

Tutorial 5

1.1 $\frac{2x+4}{2y-1}$ 1.2 $\frac{-(y^2+2x)}{2xy+3}$ 1.3 $\frac{y\sin(xy)-2xy-1}{x^2-x\sin(xy)-5}$ 1.4 $\frac{\ln 5+3y}{1-3x}$

2.2 a) $\frac{1}{4\sqrt{x^3}}$ b) $-25(\sin 5t + \cos 5t)$ c) $2\ln(x-1) + \frac{2x}{x-1} + \frac{x(x-2)}{(x-1)^2}$

2.3.1 $1 + \frac{3}{x} - \tan x - \frac{1}{2x} - \frac{2x}{x^2+5}$ 2.3.2 $y\left[\frac{1}{2x} - \frac{8}{2x+1} - 6\cot 3x\right]$

2.3.3 $y[\cos x\ln\cos x - sin^2 x\sec x]$ 2.3.4 $x^{(1+\ln x)}\left[\frac{2\ln x+1}{x}\right]$

Tutorial 6

1.1. Max $(1,5)$ b) max $(-2,3)$ min $(0,5)$ c) max $(0,0)$ $\min\left(\sqrt{3},-9\right)$ and $\left(-\sqrt{3},-9\right)$

1.2. a) i) 20 ii) 4,5 b) i) 9 ii) -3 $t=1$ $t=3$

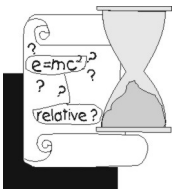
1.3 a) 0.347 b) 0,631 c) -2,457 d) 0.739 1.4 10,196 2. $19\times 10^{-5}u/s$

5. $E = -2NK\pi f\cos(2\pi f t)$ 6. $45(6+t)^{-2}$

LEARNING UNIT 3

LEARNING OUTCOME 3

INTEGRATION



Approximate study time

Two weeks

Intended Learning Outcomes

After completing this learning unit you should be able to:

- State the relationship between integration and differentiation
- evaluate indefinite integrals using basic integration rules
- integrate simple functions (polynomials, exponential and trigonometric functions)
- evaluate definite integrals
- use the definite integral to determine the area of a region enclosed by curves
- apply Simpson's rule to approximate definite integrals

CONTENTS

3.1 Integration

3.1.1 The indefinite integral as the anti-derivative and rules of integration (including method of substitution).

3.1.2 Integration involving trigonometric functions

3.1.3 Definite integrals

3.1.4 Area of a region enclosed by curves

3.1.5 Approximation of definite integrals (Simpson's rule)

3.1 Integration

3.1.1 The indefinite integral as the anti-derivative and rules of integration (including method of substitution)

Study the indicated material in the prescribed textbook (Maths 1 notes). p 65-66

Study the integration notation and go through examples.

Know standard forms. p 66

Study concepts on integration by substitution and do examples. p 67

Exercise 3.2.1

3.1.2 Integration involving trigonometric functions

Study the indicated material in the prescribed textbook (Maths 1 notes). p 69

Exercise 3.2.2

3.1.2 Definite integrals

Study the indicated material in the prescribed textbook (Maths 1 notes). p 72

Study concepts of the definite integral and do examples.

Exercise 3.**3**

3.1.3 Area of regions enclosed by curves

Study the indicated material in the prescribed textbook (Maths 1 notes). p 73-76

Study how to find the area enclosed by a curve and x-axis.

Study how to find the area enclosed by two curves.

Exercise 3.4.1 and 3.4.2

3.1.4 Approximation of Definite Integrals(Simpson's rule)

Study the indicated material in the prescribed textbook (Maths 1 notes). p 76

Read the theory and know the formula for Simpson's rule.

Know how to use Simpson's rule to approximate the area under a curve.

Exercise 3.4.3

UNIT 3: Tutorial questions

Tutorial 7

3.1 Evaluate the following integrals

a) $\int x^7 \; dx$

b) $\int 5x^4 \; dx$

c) $\int 0.6y^5 \; dy$

d) $\int 6\sqrt[3]{x} \; dx$

e) $\int 9R^{-4} \; dR$

f) $\int \frac{4}{\sqrt{x}} \; dx$

g) $\int (x^2 - x^5) \, dx$

h)

i) $\int x(x-2)^2 \; dx$

j) $\int \sqrt{x}(x^2 - x) \; dx$

k) $\int \frac{3x^2-4}{x^2} \; dx$

l) $\int (x^4+3)^4(4x^3) \; dx$

m) $\int \sqrt{8x+1} \, dx$

n) $\int \frac{x \, dx}{\sqrt{6x^2+1}}$

o) $\int \frac{dV}{(0.3+2V)^3}$

p) $\int \sec^2 (x+2) \, dx$

q) $\int \sqrt{\tan x} \sec^2 x \, dx$

r) $\int \frac{8 \sin t}{\sqrt{5-4\cos t}} \; dt$

s) $\int 3\cos^2 x \sin x \, dx$

s) $\int (1-\sin 2t)^{3/2} \cos 2t \, dt$

t) $\int \frac{8 \, dx}{2x+3}$

u) $\int e^{\sin x} \cos x \, dx$

v) $\int \frac{\cos x}{(2+\sin x)^2} \; dx$

w) $\int 2x \, e^{-x^2} dx$

x) $\int \frac{x \, dx}{4x^2+1}$

y) $\int \frac{2 \, dy}{y \ln y}$

z) $\int_0^{1/2} \frac{dx}{(2x+1)^3}$

aa) $\int_1^4 \frac{dy}{2\sqrt{y}(1+\sqrt{y})^2}$

bb) $\int_0^1 \sqrt{5x+4} \, dx$

cc) $\int_0^{\pi/6} \frac{\sin 2x}{(1-\cos 2x)^3} \; dx$

dd) $\int_0^{\sqrt{7}} x(x^2-1)^{1/3} \, dx$

ee) $\int_{-\sqrt{7}}^0 x(x^2-1)^{1/3} \, dx$

ff) $\int_{\pi/6}^{\pi/3} (1-\cos 3x) \sin 3x \, dx$

gg) $\int_{-\pi/4}^{\pi/4} \tan^2 x \sec^2 x \, dx$

hh) $\int_0^4 \frac{2x}{x^2+9} \; dx$

ii) $\int_0^{ln2} e^{-x} \, dx$

jj) $\int_1^4 \frac{e^{\sqrt{x}}}{2\sqrt{x}} \; dx$

Tutorial 8

1. Sketch the given functions and determine the area:
   a) Between the X-axis and $y = x^2 + 1$ on the interval $[0; 1]$
   b) Between Y – axis and $x = y^2 + 1$ on the interval $[-2; 2]$
   c) Between the X-axis and $y = \sin 2x$ on the interval $\left[0; \frac{\pi}{4}\right]$

d) Between the X – axis and $y = \cos x$ on the interval $\left[0; \frac{\pi}{2}\right]$

e) Between X-axis and $y = x^3$ on the interval $[-1; 1]$

f) Of the region enclosed by $y = x^2$ and $y = \sqrt{x}$

g) Of the region bounded by $y = 2x^2 + 10$ and $y = 4x + 16$

h) Of the region bounded by $y = 2x^2 + 10, y = 4x + 16, x = -2$ and $x = 5$

i) Of the region enclosed by $x = \frac{1}{2}y^2 - 3$ and $y = x - 1$

j) Of the region bounded by $x = -y^2 + 10$ and $x = (y - 2)^2$

2. Approximate the value of each of the given integrals by use of Simpson's rule, using the given value of n.

a) $\int_0^8 x^{1/3}\, dx,\ n = 2$

b) $\int_1^4 \left(2x + \sqrt{x}\right) dx,\ n = 6$

c) $\int_0^5 \sqrt{25 - x^2}\, dx,\ n = 4$

d) $\int_0^{2.4} \frac{dx}{\left(4+\sqrt{x}\right)^{3/2}},\ n = 8$

e) $\int_1^5 \frac{dx}{x^2+x},\ n = 10$

3. The time rate of change of electric current in a circuit is given by $\frac{di}{dt} = 4t - 0.6t^2$. Find an expression for the current as a function of time if $i = 2\ A$ when $t = 0\ s$.

4. The rate of change of the temperature $T\ in\ °C$ from the centre of blast furnace to a distance r in meters from the centre is given by $\frac{dT}{dr} = -4500(r + 1)^{-3}$. Express T as a function of r if $T = 2500°C$ for $r = 0$.

5.The displacement, s, of an object is given by $s = \int_0^3 (t^4 + t)dt$. Evaluate s.

6. The time taken, t hrs, for a vehicle to reach a speed of 120km/h with an initial speed of 80km/h is given by $t = \int_{80}^{120} \frac{dv}{600-3v}$ where v is velocity (km/h). Determine t.

ANSWERS: Tutorial 7

3.1.

a) $\frac{x^8}{8} + C$

b) $x^5 + C$

c) $0.1y^6 + C$

d) $\frac{9}{2}\sqrt[3]{x^4} + C$

e) $\frac{-3}{R^3} + C$

aa) $\frac{1}{6}$

bb) $\frac{38}{15}$

cc) $-\frac{1}{16}$

dd) $3{,}71352$

ee) $-3{,}71352$

f) $8\sqrt{x} + C$

g) $\frac{x^3}{3} - \frac{x^6}{6} + C$

h) $\frac{x^4}{4} - \frac{4}{3}x^3 + 2x^2 + C$

i) $\frac{2}{7}x^{7/2} - \frac{2}{5}x^{5/2} + C$

j) $3x + \frac{4}{x} + C$

k) $\frac{(x^4+3)^5}{5} + C$

l) $\frac{1}{12}(8x+1)^{3/2} + C$

m) $\frac{1}{6}\sqrt{6x^2+1} + C$

n) $\frac{-1}{4(0.3+2V)^2} + C$

o) $\tan(x+2) + C$

p) $\frac{2}{3}(\tan x)^{3/2} + C$

q) $4\sqrt{5 - cost} + C$

r) $-cos^3 x + C$

s) $\frac{-1}{5}(1 - \sin 2t)^{5/2} + C$

t) $4\ln|2x+3| + C$

u) $e^{\sin x} + C$

v) $-\frac{1}{2+\sin x} + C$

w) $-e^{-x^2} + C$

x) $\frac{1}{8}\ln|4x^2+1| + C$

y) $2\ln|\ln y| + C$

z) $\frac{3}{16}$

ff) $\frac{1}{2}$

gg) $\frac{2}{3}$

hh) 1.0217

ii) $\frac{1}{2}$

jj) 4.671

Tutorial 8

1. a) $\frac{4}{3}$

b) $\frac{28}{3}$

c) $\frac{1}{2}$

d) 1

e) $\frac{1}{2}$

f) $\frac{1}{3}$

g) $\frac{64}{3}$

h) $\frac{142}{3}$

i) 18

j) $\frac{64}{3}$

2.(a) 11.133 (b) 19.67 (c) 19.27 (d) 0.2154 (e) 0.5114

3. $i = 2t^2 - 0{,}2t^3 + 2$

4. $T = 2250(r+1)^{-2} + 250$

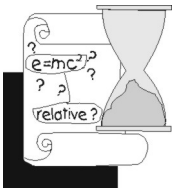
5. 53.1

6. 0.135

LEARNING UNIT 4

LEARNING OUTCOME 4

VECTORS



Approximate study time

Two weeks

Intended Learning Outcomes

After completing this learning unit you should be able to:

- perform elementary operations with vectors.
- write a vector as a linear combination of standard unit vectors.
- determine the magnitude of a vector.
- Resolve a vector into x- and y-components
- use vectors to solve problems involving force, velocity, etc-.
- analyse vectors in space (parallel vectors, collinear points).
- determine scalar products (dot products) of two vectors in the plane or in space.
- use the dot product to find the angle between two vectors.
- determine vector (cross) products in space.
- Determine the moment of a vector

CONTENTS

4.1 Elementary operations with vectors

4.2 Magnitude of a vector, Unit and parallel vectors

4.3 Resolution of vectors

4.4 Unit vector and direction vectors

4.5 Multiplication of a vector by a scalar, parallel vectors & addition and subtraction of vectors

4.6 Scalar (dot) product and the angle between two vectors and work done by forces

4.7 Cross product and the moment of a vector

4.1 Elementary operations with vectors

Study the indicated material in the precsribed textbook (maths 1 notes). p 84

4.2 Magnitude of a vector

Study the indicated material in the Prescribed textbook (Maths 1 notes). p 84

Exercise 4.3.3

4.3 Resolution of vectors

Study the concept of resolving vectors into components from the prescribed textbook (maths 1 notes) and study the examples. p 85

Do exercise 4.3.4

4.4 Unit vector and direction vectors

Study the material in the prescribed textbook (Maths 1 notes) and go through the examples. p 87

Exercise 4.3.4, 4.3.5 & 4.3.6

4.5 Multiplication by a scalar, parallel vectors & addition and subtraction of vectors

Study the material on pages 88- 89 and go through the examples.

Exercise 4.3.9 & 4.3 10

4.6 Scalar (dot) product and the angle between vectors and work done by forces.

Study the indicated material in the prescribed textbook (Maths 1 notes). p 89-91

Study concepts on scalar products of vector and the and the concept of the angle between two vectors and do examples.

Do examples on work done by forces.

Do exercises 4.3.11, 4.3.12 & 4.3.13

4.7 Cross Products and the moment of a vector

Study the indicated material in the Prescribed textbook (Maths 1 notes). p 91-92

Study concepts on cross products and do examples.

Exercise 4.3.14 and 4.3.15

Unit 4. Tutorial Questions

Tutorial 9

1 Determine the magnitude of each of the following vectors.

$\boldsymbol{m} = 4\boldsymbol{i} - 3\boldsymbol{j} + \boldsymbol{k}$ $\boldsymbol{n} = -\boldsymbol{i} + 5\boldsymbol{j} - 2\mathrm{k}$

2 Obtain the unit vectors parallel to each of the following vectors

$\boldsymbol{a} = 5\boldsymbol{i} + 6\boldsymbol{k}$ $\boldsymbol{b} = -2\boldsymbol{i} + 6\boldsymbol{j} - 6\boldsymbol{k}$ $\boldsymbol{c} = 3\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$

3.1.A force of 1200 N is applied to a body at an angle of $45°$ to the horizontal. Determine the vertical and horizontal components of the force.

3.2. The following are the magnitudes and the directions of given vectors. Determine their x- and y-components.

3.2.1. 17.8km, $\theta = 75°$ 3.2.2 75m ,$\theta = 98°$ 3.2.3. 124.8, $\theta = 195°$

3.2.4. 46.5km, $\theta = 320°$

4. Determine the dot products of the following pairs of vectors listed below:

4.1 $\boldsymbol{a} = \boldsymbol{i} - 3\boldsymbol{j} + \boldsymbol{k},\ \boldsymbol{b} = 2\boldsymbol{i} + \boldsymbol{j} - 2\boldsymbol{k}$

4.2 $\boldsymbol{a} = 3\boldsymbol{i} + 2\boldsymbol{j},\quad \boldsymbol{b} = 4\boldsymbol{i} + 6\boldsymbol{k}$

4.3 $\boldsymbol{a} = -3\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k},\ \boldsymbol{b} = 4\boldsymbol{i} - 3\boldsymbol{j} + \boldsymbol{k}$

5. Calculate the angle between each pair of vectors listed in problem 4 above.

6.a)Determine the value(s) of x if the angle between the vectors $\boldsymbol{a} = 3\boldsymbol{i} - 4\boldsymbol{j} + 2x\boldsymbol{k}$ and $\boldsymbol{b} = -2\boldsymbol{i} + 3\boldsymbol{j} + x\boldsymbol{k}$ is $90°$.

b) If the two vectors $\boldsymbol{a} = 2x\boldsymbol{i} - \boldsymbol{j} + 2\boldsymbol{k}$ and $\boldsymbol{b} = \boldsymbol{i} + \boldsymbol{x}j - \boldsymbol{k}$ are perpendicular, determine the value(s) of x.

7. Determine the work done by a force, F, in moving an object through a displacement r in each of the following:

7.1 $\boldsymbol{F} = 10\boldsymbol{i} + 2\boldsymbol{j} - 3\boldsymbol{k},\ \boldsymbol{r} = \boldsymbol{i} - \boldsymbol{j} - \boldsymbol{k}$ 7.2 $\boldsymbol{F} = 3\boldsymbol{i} - \boldsymbol{j} + 2\boldsymbol{k},\quad \boldsymbol{r} = 2\boldsymbol{i} - 3\boldsymbol{j} + \boldsymbol{k}$

8 A force $\boldsymbol{F} = 5\boldsymbol{i} + 3\boldsymbol{j} - 2\boldsymbol{k}$ is applied to move an object which moves from $A = (1, 1, 1)$ to $B = (5, -1, 2)$. Determine the work done by the force.

Tutorial 10

1.Find the cross products of the pair of vectors listed below:

$\mathbf{1.1}\ \boldsymbol{a} = \boldsymbol{i} + \boldsymbol{j} + \boldsymbol{k},\ \boldsymbol{b} = \boldsymbol{i} - \boldsymbol{j} + \boldsymbol{k}$

$\mathbf{1.2.}\ \boldsymbol{a} = 3\boldsymbol{i} + 2\boldsymbol{j},\quad \boldsymbol{b} = 4\boldsymbol{i} + 6\boldsymbol{k}$

$\mathbf{1.3.}\ \boldsymbol{a} = 5\boldsymbol{i} + \boldsymbol{j} + 3\boldsymbol{k},\ \boldsymbol{b} = 2\boldsymbol{i} - 3\boldsymbol{j} - 4\boldsymbol{k}$

2. Determine the moment vector, M, about the origin, O, of a force $\boldsymbol{F} = 2\boldsymbol{i} - \boldsymbol{j} + \boldsymbol{k}$, passing through the point with position vector $\boldsymbol{r} = -\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$.

3. Given that $a = 5i + 2j - k$ and $b = i - 3j + k$ determine $(a + b) \times (a - b)$

4. Given the position vectors $\overrightarrow{OA} = 2i - j + 3k$ $\overrightarrow{OB} = 3i + 2j - 4k$ $\overrightarrow{OC} = -i + 3j - 2k$, Determine

a)the vector $\overrightarrow{AB}$

b) the vector $\overrightarrow{BC}$

c) the vector product $\overrightarrow{AB} \times \overrightarrow{BC}$

5. A force $F = 5i - j - k$ passes through point A whose position vector is $a = 3i + j + 8k$. Determine the vector moment, M, of F about the point B with position vector $b = 2i - 3j + 7k$. Also determine the magnitude of M.

ANSWERS

Tutorial 9

1. $|\boldsymbol{m}| = \sqrt{26}$ $|\boldsymbol{n}| = \sqrt{30}$

2. $\hat{\boldsymbol{a}} = \frac{1}{\sqrt{61}}(5i + 6k)$ $\hat{\boldsymbol{b}} = \frac{1}{\sqrt{76}}(-2i + 6j - 6k)$ $\hat{\boldsymbol{c}} = \frac{1}{\sqrt{14}}(3i + 2j - k)$

3.1 $F_x = 848.53$ $F_y = 848.53$

3.2.1. $\begin{pmatrix} 4.607 \\ 17.193 \end{pmatrix}$ 3.2.2 $\begin{pmatrix} -10.438 \\ 74.270 \end{pmatrix}$ 3.2.3 $\begin{pmatrix} -120.548 \\ -32.301 \end{pmatrix}$ 3.2.4 $\begin{pmatrix} 35.621 \\ -29.890 \end{pmatrix}$

4.1 -3 4.2 12 4.3 -19

5.1 $\theta = 107{,}55°$ 5.2 $\theta = 62.5°$ 5.3 $\theta = 174{,}79°$

6.a) ± 3 6. b) 2

7.1 $11\,J$ 7.2 $11J$ 8. $12J$

Answers: Tutorial 10

1.1 $2\boldsymbol{i} + 0\boldsymbol{j} - 2\boldsymbol{k}$ 1.2 $12\boldsymbol{i} - 18\boldsymbol{j} - 8\boldsymbol{k}$ 1.3 $5\boldsymbol{i} + 26\boldsymbol{j} - 17\boldsymbol{k}$ 2 $\boldsymbol{i} - \boldsymbol{j} - 3\boldsymbol{k}$

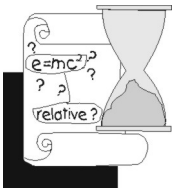
3. $2i + 12j + 34k$ 4. a) $i + 3j - 7k$ b) $-4i + j + 2k$ c) $13i - 30j + 13k$

5. $-3i + 6j - 21k$ 22.045

LEARNING UNIT 5

LEARNING OUTCOME 5

COMPLEX NUMBERS



Approximate study time

Two weeks

Intended Learning Outcomes

After completion of this learning unit you should be able to:

- perform operations (addition/subtraction/multiplication/division) with complex numbers in rectangular form
- represent a complex number on the Argand diagram.
- convert a complex number from the rectangular form to the polar form and vice versa
- use De Moivre's theorem to find products, quotients and powers (including roots) of complex numbers in polar form

CONTENTS

5.1 Representation and operations with complex numbers in rectangular form.
5.2 Equality of complex numbers
5.3 Argand diagram and polar form of complex numbers
5.4 De Moivre's theorem
5.5 Using De Moivre's Theorem to calculate powers and roots of real and complex numbers

5.1 Representation and operations with complex numbers in rectangular form

Study the indicated material in the prescribed textbook (Maths 1 notes). p 95-97
Study the section on representation of complex numbers in rectangular (standard) form. Go through the examples.
Study the arithmetic of complex numbers (addition, subtraction, multiplication and division) and do examples.

Exercises 5.2, 5.3, 5.4 & 5.5

5.2 Equality of complex numbers

Study the indicated material in the prescribed textbook (Maths 1 notes). p 98

Go through the examples.

Exercise 5.6

5.3 Argand diagram and polar form of complex numbers

Study the indicated material in the prescribed textbook (Maths 1 notes). p 98

Study how to represent a complex number on the complex plane.

Study polar form representation of a complex number

and do examples.

Exercise 5.

5.4 De Moivre's Theorem (powers and roots)

Study the material in the prescribed textbook (Maths 1 notes). p 101 – 102

Follow the examples of De Moivre's theorem and use it to calculate roots of real and complex numbers.

Exercise 5.8 and 5.9

Unit 5. Tutorial Questions

Tutorial 11

1. Write the following in j-notation:

1.1 $\sqrt{-4}$ 1.2 $\sqrt{(-9)(-4)}$ 1.3 $\sqrt{-9}\sqrt{-4}$

2. Solve the following quadratic equations. Write your answers in the form $a \pm bj$

2.1) $x^2 + 6x + 10 = 0$ 2.2) $2x^2 - 4x + 4 = 0$ 2.3) $3x^2 - 2x + 1 = 0$

3. Find the complex conjugate of:

3.1 $z = 5 + j4$ 3.2 $z = -2 + j$ 3.3 $z = -3 - 2j$ 3.4 $z = 1 - 5j$

4. Given that $z_1 = -2 - 7j,\ z_2 = 4j$, determine the following:

4.1 Re(z_1) 4.2 Im($\bar{z}_2$) 4.3 Re(z_2) 4.4 Im($\bar{z}_1$)

5. Determine in standard form: $\overline{1 - \mathrm{j}} + \sqrt{-36} - j^5 + j^{-3} - \mathrm{Im}(-3 - 2j)$

6. Simplify leaving the answer in standard form / rectangular form:

6.1 $(2 + 3j) + (-1 + j4)$ 6.2 $(-1 - j) - (2 - 3j)$ 6.3 $-2\left(\frac{1}{4} - j\frac{5}{8}\right)$

6.4 $(2 + 3j)(-1 + 4j)$ 6.5 $j^{25} + j^{-2}$ 6.6 $\frac{(j+2x)}{j}$

7. Determine the roots of:

7.1 $x^3 + 8 = 0$ 7.2 $x^2 - x + 1 = 0$ 7.3 $x^3 - 1 = 0$

8. Solve for x and y if:

8.1 $(1 + j)(2 - yj) = x + j4$ 8.2 $2x - 6xj^3 - 3j^2 = yj - y + 7j^5$

8.3 $\frac{j+2x}{j} = y - j^5$ 8.4 $(x - 2y)(3 + j4) - x(2 - j) = -4 + 2j$

Tutorial 12

1. Convert the following complex numbers to polar form

1.1 -1 + $\sqrt{-3}$ 1.2 3.514 - 7.256j 1.3 $\sqrt{3} + j\sqrt{3}$

2. Simplify giving the answer in standard form:

2.1 $j^6 + \sqrt{-9} - 3cis\frac{\pi}{4} + \overline{3 - 2j}$ 2.2 $4cis(-30^\circ) + 2cis90^\circ$

2.3 $2\angle 60^\circ \cdot 3\angle 40^\circ \cdot \angle 90^\circ$ 2.4 $\frac{3\angle 16^\circ \cdot (2\angle 40^\circ)}{\angle 70^\circ}$

3. Express each of the given numbers in exponential form and standard form:

3.1 $3\,(\cos 60^0 + j\sin 60^0)$ 3.2 $2{,}56(\cos 125{,}2^0 + \mathrm{j}\sin 125{,}2^0)$

3.3 $375{,}5[\cos(-95{,}46^0) + j\sin(-95{,}46^0)]$

4. Use De Moivre's Theorem to determine: (Give the answer in rectangular form)

4.1 $(-1 - j)^{12}$ 4.2 $(2 + 2j)^{-6}$ 4.3 $\frac{(-1+\sqrt{3}j)^3}{\sqrt{2}\angle(-30^\circ)}$

4.4 $\frac{3\angle 16^\circ \cdot (2\angle 40^\circ)^3}{\angle 70^\circ}$ 4.5 $\frac{(3-j4)^2(4cis30^\circ)}{10cis\frac{\pi}{2}}$

5. Given that the voltage in a given circuit is 8,375 – 3,140j V and the impedance is 2,146 – 1,114j ohms, find the magnitude and phase angle of the current.

6. In an electric circuit, the admittance is the reciprocal of the impedance. If the impedance is $2800 - 1450j$ ohms in a certain circuit, find the exponential form of the admittance.

7. An a.c. voltage, $V = 2.0 + 6.0j$ volts is applied across an impedance, $Z = (8 + j)ohms$. Find the current, i in polar form through the circuit given that $i = \frac{V}{Z}$

8. A system has the following close-loop transfer function $G = \frac{K(Z^4+16)}{Z^3+64}$ where K is a non-zero constant.
The poles of the system occur where the denominator of the transfer function is zero, that is $Z^3 + 64 = 0$. The zeros of the system occur where the numerator of the transfer function is zero, that is $Z^4 + 16 = 0$.
   a) Determine the poles of the system and label them on an Argand diagram.
   b) Determine the zeros of the system and label them on an Argand diagram.

9. Determine all the roots of the given equations:
9.1 $x^3 - 8 = 0$ 9.2 $x^2 + 2 = 0$ 9.3 $x^4 - 1 = 0$

ANSWERS

Tutorial 11

1.1 2 j 1.2 6 1.3 - 6

2.1. $-3 \pm 2j$ 2.2. $1 \pm j$ 2.3. $\frac{1\pm\sqrt{2}}{3}$

3.1 $5 - 4j$ 3.2 $-2 - j$ 3.3 $-3 + 2j$ 3.4 $1 + 5j$
4.1 -2 4.2 -4 4.3 0 4.4 7 5. 7j+3

6.1 $1 + 7j$ 6.2 $-3 + 2j$ 6.3 $-\frac{1}{2} + j\frac{5}{4}$ 6.4 $-14 + 5j$

6.5 $-1 + j$ 6.6 $1 - 2xj$

7.1 $x = -2;\ \ x = 1 + 1.732j;\ \ x = 1 - 1.732j$
7.2 $x = 0.5 + 0.866j;\ \ x = 0.5 - 0.866j$

7.3 $x = 1, x = -0.5 + 0.866j;\ \ x = -0.5 - 0.866j$

8.1 $x = 0\,,\ y = -2$ 8.2 $x = 0.5\ \ y = -4$ 8.3 $x = 0.5\,,\ y = 1$
8.4 $x = 2\,,\ y = 1$

Tutorial 12

1.1 $z = 2Cis\ 120^{\circ}$ 1.2 $z = 8.06Cis\ 64.16^{\circ}$ 1.3 $z = 2.449Cis\ 45^{\circ}$

2.1 $-0.121 + 2.88j$ 2.2 $3.464 + 0j$ 2.3 $-5.909 - 1.042j$

2.3 $5.822\ +\ 1.452j$

3.1 $3e^{\frac{\pi j}{3}} = 1{,}5 + 2{,}598j$ 3.2 $2.56e^{0.6956\pi j} = -1{,}476 + j2{,}092$

3.2 $375.5e^{-0{,}5303\pi j} = -35{,}729 - 373{,}796j$

4.1 - 64 4.2 $0 + 0.002j$ 4.3 $4.899 + 2.828j$

4.4 $9.762 + 21.925j$ 4.5 $-5.189 - 8.548j$

5. $5.863,\ 4.336^{\circ}$ 6. $3{,}1714 \times 10^{-4}e^{j4{,}778x\ 10^{-1}}$ 7. $0{,}030cis64{,}4°$

8. a) $2 \pm 2\sqrt{3}\ ,\ \ -4.$ b) $\sqrt{2}(1 \pm j),\ \ -\sqrt{2}(1 \pm j)$

9.1 $2, -1 \pm \sqrt{3}j$ 9.2 $\pm\sqrt{2}j$ 9.3 $\pm 1,\ \ \pm j$